

Appendices: proposed changes for review

Toward Personalized Hebrew ASR · Dolev Abudi and Hadas Yonat · 7 October 2026

This file is the whole paper with the changes proposed to the **appendices** marked against the version on `main`. Only the appendices change: the main text and Limitations are already merged. **The appendices start on page 8 of this file.**

How to read it: ~~red, struck-out~~ is removed, blue, underlined is added, and black text is unchanged.

What changed, and why

- **A, Data Preparation.** One finding added to *Two test sets*: the quality score predicts arm A’s errors about as well as arm B’s (Spearman -0.62 and -0.63 per segment, recomputed over the 65,990 segments), so filtering on it is not specific to arm B. It answers the obvious question about a score from a Whisper-family aligner. The fourth condition of the training filter is rewritten shorter, with the same meaning.
- **B, Training Details.** *Tuning results* becomes prose instead of a bullet list, with the same content. The tuning procedure is one paragraph. All semicolons removed (in the text, three cells of Table 1 and two captions). “meetings” becomes “sessions”, as in the main text.
- **C, Implementation Details.** Two methodological points the paper did not state: the control never trains on a session in which a speaker it is scored on is tested or validated, and decoding is greedy with a loop guard (a hypothesis whose compression ratio exceeds 3.0 is decoded again with no 6-token sequence repeated, the same for every model, as in `evaluate.py`). The bootstrap details are replaced by a pointer to Appendix E, where they are given in full.
- **D, Additional Results.** The decision rule is kept, because it was fixed before the runs and the main text does not state it, and shortened. Its intervals were checked against `curve_v4.csv`. A heading for the 360-minute paragraph, “session-disjoint”, one sentence shorter in *Reliability*, and a caption without semicolons.
- **E, Significance Testing.** All semicolons removed. The counts were checked: 8 and 16 significantly positive pairs, none negative.
- **F and G** are unchanged.

Also checked: the rank-32 table matches `capacity_v4.csv`, and the 0.381 against 0.391 learning-rate check and the 0.83–1.11 passes match the run 4 results.

Toward Personalized Hebrew ASR: Adapting a General Model to a Single Speaker

Dolev Abudi

Tel Aviv University
dolevabudi@mail.tau.ac.il

Hadas Yonat

Tel Aviv University
hadasyonat@mail.tau.ac.il

Abstract

Automatic speech recognition (ASR) systems are built for general-purpose use and evaluated by error rates averaged across many speakers, which says little about how well they serve any one person. Yet in many deployments it is a single speaker’s accuracy that matters. We ask how to personalize a Hebrew ASR model to one speaker under a small data and compute budget. On Knesset committee recordings, we first measure, for each of 267 speakers, the error of a general Whisper model and of its Hebrew fine-tune. The fine-tune lowers every speaker’s error but leaves speakers as unequal as before, and group membership explains little of who is served well. We then adapt the Hebrew model to 12 speakers with LoRA, and compare each speaker’s own adapter with a control adapter trained on the same amount of speech from other speakers. With 80 minutes of a speaker’s speech, the own adapter lowers the word error rate by a median 12%, but the control already lowers it by 10%, so the personal gain is only 2–3%. With up to 24 hours of a speaker’s speech, the own adapter keeps improving, but the control improves as fast, and the personal gain does not grow. Most of what adaptation buys is shared across speakers, so a claim of personalization needs a control of this kind. Our code is available at <https://github.com/hadasy-tau/deep-learning-project>.

1 Introduction

Modern ASR models such as Whisper (Radford et al., 2023) are evaluated by error rates averaged over many speakers, an average that can hide the speakers a model serves poorly. Yet many deployments serve one particular person, such as a Knesset member transcribed every day or the single user of a voice assistant. What matters to them is how well the model recognizes their own voice, which calls for personalization: adapting a general model to one speaker with a small amount of their own speech.

We study personalization for Hebrew and ask three questions: does a Hebrew fine-tune serve all speakers equally, what does a speaker’s own speech add over the same amount of other speakers’ speech, and does that personal gain grow with more of the speaker’s data? We answer them on Knesset committee recordings, which the Hebrew model was not trained on, and compare each speaker’s own LoRA adapter with a control trained on the same amount of other speakers’ speech. Our contributions are threefold. First, a speaker-level error map, the first for Hebrew ASR, shows that the Hebrew fine-tune helps every speaker but leaves them as unequal as before. Second, a controlled comparison shows that at 80 minutes a speaker’s own audio adds only 2–3% over the control. Third, a data curve to 24 hours of one speaker’s speech shows that this personal margin does not grow. Before scoring on the test sets, we tuned the learning rate, LoRA rank, Whisper’s dropout and augmentation in 64 runs on validation data, and at 24 hours we also tried a four times larger adapter. Training is cheap: an adapter on 80 minutes of speech trains in a few minutes on a standard GPU.

2 Related Work

Averages hide who is served. Whisper (Radford et al., 2023) is much weaker on languages with little training data, Hebrew among them. ivrit.ai closed this gap by fine-tuning it on Hebrew, mostly Knesset plenary speech (Marmor et al., 2025), the starting point for our work. Both models are evaluated by error rates averaged over large test sets, never per speaker or subgroup.

Gaps within a language. Within a single language, too, ASR serves some people worse than others. Error rates differ by race (Koenecke et al., 2020), and Tatman (2017), Feng et al. (2021) and Liu et al. (2022) find gaps by gender, dialect, age and accent. These studies compare groups, and we

know of no such analysis for Hebrew. We provide one, for each speaker and for subgroups that share gender, age, origin or speaking rate.

Adapting a model to one speaker. Adapting a recognizer to its user is an old idea in speech recognition (Gales, 1998). With modern models, fine-tuning on a small amount of a person’s own speech can greatly reduce their errors, but this has mostly been studied for atypical speech, such as speech impairments (Green et al., 2021) and strong accents (Shor et al., 2019). Training a full model for every speaker is expensive, so methods such as LoRA (Hu et al., 2022) train only a small number of added parameters, and have also been used with Whisper (Liu et al., 2024). For typical speakers, Weninger et al. (2019) report gains that keep growing up to 20 hours of a speaker’s speech, but measure them against the unadapted model only, so they also include what any in-domain speech would buy. We separate the two with a control trained on the same amount of other speakers’ speech.

3 Experimental Setup

3.1 Overview

Our study has two experiments. In the first, a general Whisper model and its Hebrew fine-tune (Section 3.3) transcribe the same speech, and we measure the error of each speaker and subgroup under both, and so the gain the fine-tune brings. In the second, we select 12 speakers from this map: speakers the fine-tune helped least, speakers hard under both models, typical speakers, a low-WER speaker, and two speakers the fine-tune served well. To avoid selecting on measurement noise, we rank speakers on their earlier sessions only and hold out their newest sessions for testing. We then fine-tune the Hebrew model on each speaker’s own speech and compare it with a control adapter trained on the same amount of speech from other speakers, which separates learning the voice from learning committee speech in general.

3.2 Data

We use ivrit-ai/knesset-committees, recordings of Israeli parliamentary committee sessions with their human-written protocols, force-aligned to the audio. Each protocol segment carries the speaker’s name and an alignment-confidence score. Because the protocols are stenographic rather than verbatim, a low score may reflect a mismatch between the text and the audio. The corpus spans

November 2018 to January 2026 with 13,177 sessions and 15,466 recorded hours, of which 7,886 are aligned speech with confidence above 0.7.

3.3 Models

We compare two models with the same architecture. Arm **A** is openai/whisper-large-v3 (Radford et al., 2023), the general multilingual model, and arm **B** is ivrit-ai/whisper-large-v3 (Marmor et al., 2025), the same model fine-tuned on roughly 4,700 hours of Hebrew *plenary* speech, so our committee recordings are new to it, though many of the speakers are not (see Limitations). Every adapter in this work is trained on top of arm B. For the speaker-level analysis, arm B is served through ivrit.ai’s inference checkpoint (whisper-large-v3-turbo-ct2, with 4 decoder layers instead of 32).

3.4 Preprocessing

The released corpus needs two steps before use: Whisper takes at most 30 seconds of audio while sessions last hours, and a speaker-level analysis needs every segment linked to a verified person. We follow the procedure of VoxKnesset (Marmor et al., 2026), a 2,300-hour corpus of Knesset plenary speech with verified speakers, but not its audio, which comes from the plenary sessions arm B was fine-tuned on.

Protocol speaker names are matched exactly to the Knesset’s official roster of name variants and accepted only within the member’s term of service, and unresolved names are discarded. Each member’s birth year, gender and origin are joined through the matched identity. Each identified turn is then cut into segments of at most 30 seconds that never cross a speaker change, with the aligned protocol text as the reference. The result is 1,204,617 segments (3,841 hours) of 330 current and former Knesset members in 10,905 sessions.

Each segment carries an alignment quality score between 0 and 1 (Appendix A). Segments below 0.7 are excluded from all analyses, and training and validation data are further restricted to segments above 0.95, following ivrit.ai’s recommendation.

3.5 Speaker-Level Performance

Both models transcribe the same segments: one hour of speech per speaker, drawn round-robin over their sessions so that it spans their tenure, 65,990 segments (230 hours) from 267 speakers in all. Arm A runs through Hugging Face Inference

Providers and arm B through a RunPod endpoint with ivrit.ai’s inference checkpoint, and both are told the language is Hebrew. For each speaker and subgroup, we report WER and CER under both models, the gain $WER_A - WER_B$, and the relative gain, $(WER_A - WER_B)/WER_A$.

3.6 Personalization

Speakers and data. The sessions of each of the 12 selected speakers are split into training, validation and test sets that share no session, with the newest sessions held out for testing (Appendix D). Each speaker is trained on 5, 20 and 80 minutes of their own speech, nested and filled from their most recent training sessions first, and four speakers later on up to 1,440 minutes (Appendix B). Every model is scored on two test sets from the same held-out sessions (Appendix A): the high-quality segments (*clean test*, median 47 minutes per speaker) and all segments at quality ≥ 0.7 (*full test*, median 90 minutes).

Adapter and training. We train LoRA adapters (Hu et al., 2022) of rank 8 on the query and value projections of every attention layer of arm B (3.9M trainable parameters), with the learning rate and the other settings chosen on validation data (Appendix B), for at most 400 steps with early stopping: validation loss is measured every 20 steps on other sessions of the speaker, used for neither training nor testing, and training stops after four checks without improvement, restoring the best checkpoint. The training labels are the protocol text.

Control. An adapter trained on a speaker’s audio learns both the speaker and the committee setting: rooms, microphones, vocabulary and the protocol’s style. To separate the two, the 12 speakers are split at random into two folds of six. For each fold and budget, a *control* adapter is trained with the same recipe on the same number of minutes from the six speakers of the *other* fold. Each speaker is scored on the same test segments under their own adapter and under the control, which never heard them. We report the relative gain over arm B, $(WER_B - WER_{\text{adapted}})/WER_B$, and the **personalization**, $(WER_{\text{control}} - WER_{\text{own}})/WER_B$, which equals the own gain minus the control gain. Its 95% interval and p-value come from a paired bootstrap over test segments (Appendix E). All budgets use seed 0, and at 80 minutes seeds 1 and 2 repeat the own adapters and the controls, with the folds redrawn, to separate speaker-level effects from seed

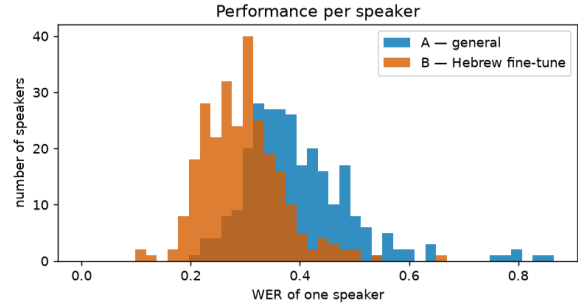


Figure 1: Per-speaker WER under arm A (general) and arm B (Hebrew fine-tune), all 267 speakers. Arm B shifts the whole distribution down without narrowing it in proportion.

variance.

4 Results

4.1 Speaker-Level Error Map

After the quality filter, the analysis covers 267 speakers and 217 hours (58,180 segments), with a corpus WER of 0.387 for arm A and 0.292 for arm B. Statistics below are over the 261 speakers with at least 20 segments, and per-speaker 95% confidence intervals come from bootstrapping each speaker’s segments (Appendix F).

The fine-tune helps every speaker but leaves them as unequal. All 261 speakers have a lower WER under arm B, and for every one of them the improvement is beyond sampling noise: the 95% bootstrap interval of their gain lies entirely above zero (Figure 10, Appendix D). Yet the spread of per-speaker WER shrinks (standard deviation 0.099 to 0.069) only in proportion to the lower average (coefficient of variation 0.25 and 0.23, Figure 1). Speakers keep their order (Spearman $\rho = 0.90$ between WER_A and WER_B), and this ranking is stable across a speaker’s recordings (Appendix D).

Group membership explains little. We compare the gain across seven groupings of speakers: gender, nationality, religion, religious orientation, country of origin, age quartile and speaking-rate tertile. The clearest difference is speaking rate (Figure 2, with the other six in Figure 12): slow speakers gain more than fast ones (27% vs. 22% of arm A’s error removed). Arab, Muslim, Druze and Bedouin speakers also gain more than Jewish speakers (29–30% vs. 24%). But group sizes are highly imbalanced (229 Jewish speakers against 5 Bedouin), and the variation within each group far exceeds the differences between group medians. A

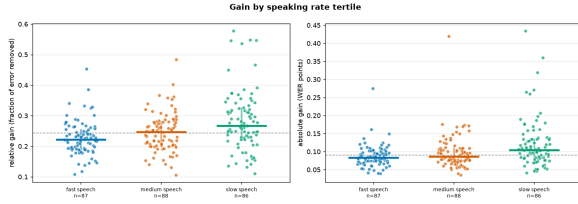


Figure 2: Gain from the Hebrew fine-tune by speaking-rate tertile, relative (left) and absolute (right). Each point is one speaker, the bar is the group median, and the dashed line is the overall median.



Figure 3: Relative WER change over arm B at 5, 20 and 80 minutes of training audio (seed 0, 12 speakers), on the clean test (left) and the full test (right). Blue is the speaker's own adapter, orange the control, trained on the same minutes from other speakers, and green the personalization (own – control). Each dot is one speaker, and bars and numbers are medians. The median personalization is the median of per-speaker differences, not the difference of the two medians.

speaker's group thus says little about how well the model serves them, which motivates adapting to individual speakers.

4.2 Adaptation Results

The 80-minute adapter improves almost every speaker. With 80 minutes and seed 0, the own adapter lowers arm B's WER by a median 12.0% on the clean test and 15.3% on the full test (Figure 3, blue). It is better than arm B for 11 of 12 speakers on the clean test (9 significantly) and for all 12 on the full test (all significantly). The other two seeds agree: over all 36 speaker–seed pairs, it beats arm B in 35 on the clean test and in all 36 on the full test, and averaged over the three seeds, the clean-test gains range from +3% to +28% (median +11.4%, Figure 4).

Personal versus shared adaptation. At 80 minutes, the median personalization over the 36 speaker–seed pairs is +2.4% on the clean test and +2.2% on the full test. With seed 0, the control, which never heard the speaker, already gains +10.1% and +14.3% (Figure 3), against the own

adapter's +12.0% and +15.3%: most of the adaptation gain is general, and the speaker's own audio adds 2–3% on top of it.

Five speakers are significant in all three seeds on the full test (Figure 5), led by 23558 at +9% to +10%, while six never are, in any seed or test. Profile does not predict the effect: speakers the fine-tune helped least and speakers hard under both models fall on both sides. A speaker's personalization varies across seeds with a median standard deviation of 2.4% on the clean test, as large as the effect, so we make per-speaker claims only where all seeds agree. Nor does the margin grow on cleaner audio: by alignment-quality band of the full test, own adapter and control gain alike on the two hardest bands (about +20%), and the median personalization is +0.5%, +4.0%, +2.5% and +2.3% from the hardest band to the cleanest.

How much speaker data is needed. The own adapter's median relative gain over arm B grows from +4.3% to +7.5% to +12.0% on the clean test at 5, 20 and 80 minutes, and from +4.2% to +9.7% to +15.3% on the full test (Figure 3), and no 5- or 20-minute adapter is the best model for any speaker (Figure 4). The control rises almost as fast, to +0.3%, +7.8% and +10.1% on the clean test and +6.7%, +10.9% and +14.3% on the full test, so what extra audio buys is mostly not specific to the speaker, and the personal margin stays between +1.6% and +3.1% on the clean test at every budget. At 5 minutes both adapters are unreliable: they are worse than arm B for some speakers, and the personal margin has opposite signs on the two test sets.

Beyond 80 minutes. To test whether personalization is limited by the amount of speaker data, we train 23558, the speaker with the largest personal effect, and 23641, a speaker outside the 12 with one of the highest WERs, on up to 1,440 minutes (24 hours) of their own speech, each budget with its own control (Figure 6, Appendix B). The own adapter keeps improving, with diminishing returns: on the full test, from +29.4% at 80 minutes to +34.3% at 24 hours for 23558, and from +17.4% to +23.4% for 23641. The control improves at least as fast (for 23558, from +19.7% to +30.2%), so the added gain comes mostly from committee speech in general, and personalization does not grow. It shrinks as a share of the gain, for 23558 from a third to an eighth, but it does not vanish: at 24 hours it is +4.1% for 23558 and +2.3% for 23641.

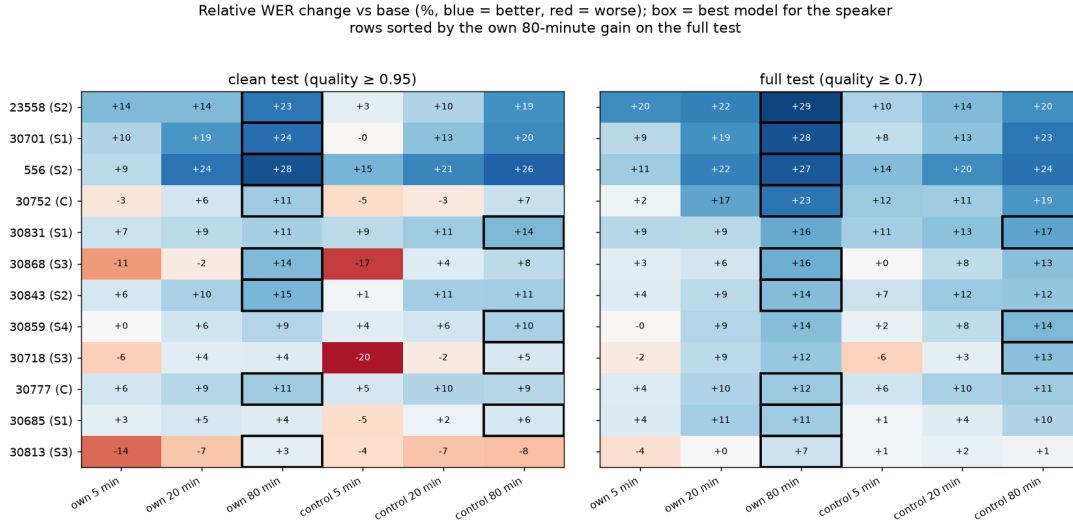


Figure 4: Relative WER change over arm B (%) for each speaker under each adapter. Blue is better, red is worse, and the box marks the speaker’s lowest WER. The 80-minute columns are averaged over three seeds. Rows are sorted by the own 80-minute gain on the full test.

Longer training does not change this: validation loss stops improving after about one pass over the data, and a larger adapter (rank 32) does not help (Appendix B). Two more speakers, trained on up to 360 minutes, show the same pattern (Figure 17).

Learning curves. Validation loss reaches its minimum at step 60–100, about two passes over the data and 38–54% below the validation loss of arm B without an adapter, and then rises as the adapter overfits (Figure 7). Every run stops early, by step 180 of 400, and the three seeds trace nearly identical curves (Figure 16, Appendix D). An adapter trains in a few minutes on a standard GPU.

5 Conclusion

The Hebrew fine-tune lowers every speaker’s error but leaves them as unequal as before, which motivates adapting to individual speakers.

Most of the adaptation gain is general. An 80-minute LoRA adapter on a speaker’s own committee speech lowers arm B’s WER by a median 12% on the clean test and 15% on the full test, and the same amount of other speakers’ audio already lowers it by 10% and 14%. Training on the speaker’s own audio adds a further 2–3% over this shared baseline, positive in all three seeds.

More audio gives more gain, but mostly not personal gain. From 5 to 80 minutes the own adapter’s gain roughly triples, and the control’s grows almost as fast. Up to 24 hours of a speaker’s

own speech, the own adapter keeps improving, and a control trained on as much audio from other speakers improves as fast: the added gain does not require the speaker’s own speech, and the personal margin does not grow with it.

Adaptation is cheap. An adapter converges in about two passes over its data and trains in a few minutes on a standard GPU. The recipe is insensitive to rank, augmentation and, at 80 minutes, the learning rate, but Whisper’s dropout prevents learning.

Since most of what adaptation buys is shared across speakers, a claim of personalization needs a control trained on other speakers’ speech. Further directions are adaptation on similar rather than random speakers, and training objectives that separate the voice from the committee setting.

Limitations

Prior exposure. Arm B was fine-tuned on Kneset plenary speech, which includes many of our speakers. Our evaluation uses committee recordings, a different setting and register, but the model may still be partly familiar with these voices.

Shared recording conditions. Train and test sets are session-disjoint, but not committee-disjoint: about 75–82% of the test speech comes from committees the same speaker also appears in during training, and so likely from the same rooms and microphones. Part of an adapter’s gain may therefore

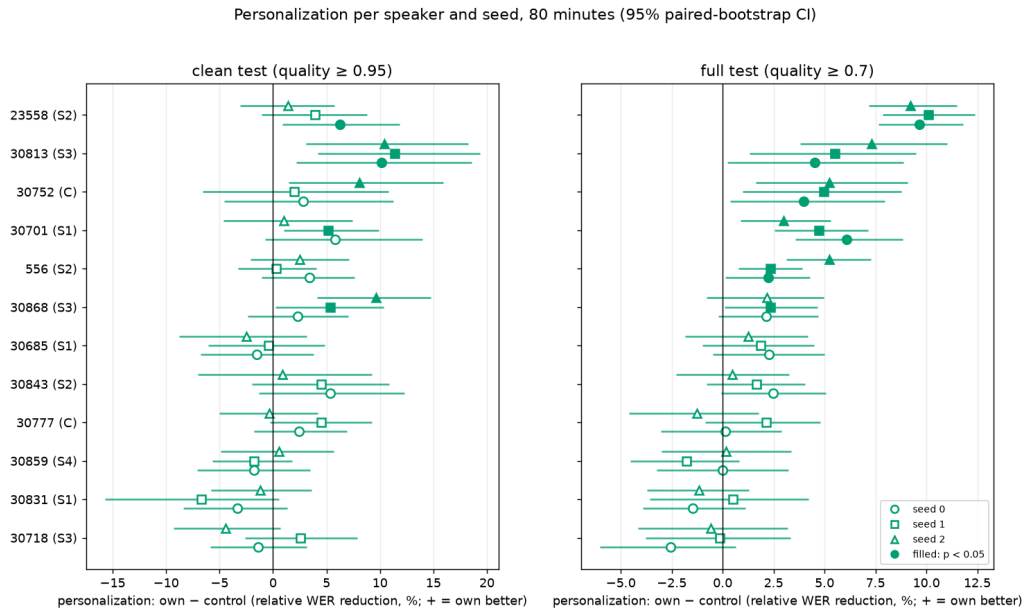


Figure 5: Personalization (own – control, relative WER reduction in %) per speaker and seed at 80 minutes, on the clean test (left) and the full test (right), with 95% paired-bootstrap intervals. Filled markers show an own adapter significantly better than the control (paired bootstrap, $p < 0.05$), hollow markers one that is not. Speakers are sorted by their mean on the full test. Profiles: S1, helped least by the fine-tune, S2, hard under both models, S3, typical, S4, low WER, and C, served well by the fine-tune.

reflect familiar recording conditions rather than the voice alone.

The reference is the protocol. Every WER is measured against the stenographic protocol, which is not a verbatim transcript: stenographers condense, rephrase and omit speech, so part of every measured error is a mismatch between protocol and audio rather than a recognition mistake. Our gains are gains in agreement with the protocol, and may partly reflect learning what the stenographer omits. We limit this by training and validating only on segments whose protocol closely matches the audio (quality ≥ 0.95 , a threshold set with the ivrit.ai team, Appendix A). That score, however, comes from a Whisper-family aligner, so a test set filtered the same way would favour speech that Whisper models already transcribe well, and we therefore score every model both on the high-quality segments and on all segments at quality ≥ 0.7 . No verbatim reference was available, so our comparisons are most reliable between models scored against the same references.

Limited compute. All experiments ran on paid cloud services (RunPod GPUs and Hugging Face inference), for about \$200 in total, which limited the models, configurations and speakers we could explore.

Tuning on validation loss. The rank, dropout, augmentation and learning rate were chosen on the validation loss of four speakers. The rejected settings, including rank 16, were never scored on the test sets.

Few speakers beyond 80 minutes. The longer budgets cover four speakers, two of them up to 24 hours, and the rank-32 check uses one seed, so the finding that the personal margin does not grow with more speaker data may not hold for every speaker.

A random control group. The control pools speakers at random, so it does not test whether *similar* speakers could replace a speaker’s own audio.

Acknowledgments

We thank the ivrit.ai team (ivrit.ai, with models and datasets at <https://huggingface.co/ivrit-ai>, and Marmor et al., 2025) for their support throughout this project. Their openness in sharing ideas, practical advice and lessons from their own work on Hebrew speech data helped shape our direction, and their open models and datasets made this study possible.

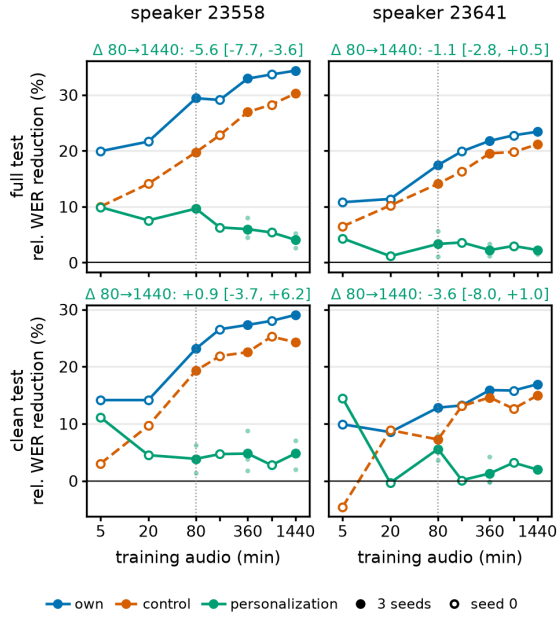


Figure 6: Relative WER reduction over arm B for the two speakers trained on up to 1,440 minutes, on the full test (top) and the clean test (bottom). Blue: the speaker’s own adapter. Orange: the control, trained on the same minutes from other speakers. Green: personalization (own – control). Filled markers are means of seeds 0–2 (small dots: the single seeds), hollow markers seed 0. The dotted line marks 80 minutes, the largest budget of the main experiment. Above each panel: the change in personalization from 80 to 1,440 minutes, with its 95% paired-bootstrap interval. At 5 minutes both adapters are unreliable (Section 4.2).

References

- Siyuan Feng, Olya Kudina, Bence Mark Halpern, and Odette Scharenborg. 2021. [Quantifying bias in automatic speech recognition](#). *Preprint*, arXiv:2103.15122.
- Mark J. F. Gales. 1998. [Maximum likelihood linear transformations for HMM-based speech recognition](#). *Computer Speech & Language*, 12(2):75–98.
- Jordan R. Green et al. 2021. [Automatic speech recognition of disordered speech: Personalized models outperforming human listeners on short phrases](#). In *Proc. Interspeech 2021*.
- Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2022. [LoRA: Low-rank adaptation of large language models](#). In *International Conference on Learning Representations*.
- Allison Koenecke et al. 2020. [Racial disparities in automated speech recognition](#). *Proceedings of the National Academy of Sciences*, 117(14):7684–7689.
- Chunxi Liu et al. 2022. [Towards measuring fairness](#)

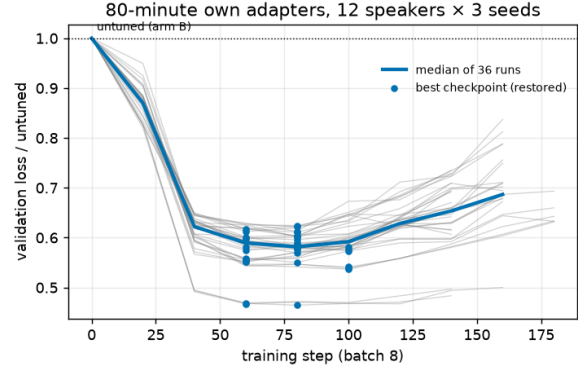


Figure 7: Validation loss relative to arm B while training the 80-minute own adapters (12 speakers $\times$ 3 seeds). Thin lines are single runs, the thick line is the median, and dots are the checkpoints early stopping restored.

in speech recognition: Casual conversations dataset transcriptions. In *Proc. ICASSP 2022*.

Wei Liu, Ying Qin, Zhiyuan Peng, and Tan Lee. 2024. [Sparsely shared LoRA on Whisper for child speech recognition](#). In *Proc. ICASSP 2024*, pages 11751–11755.

Yanir Marmor, Yair Lifshitz, Yoad Snapir, and Kinneret Misgav. 2025. [Building an accurate open-source Hebrew ASR system through crowdsourcing](#). In *Proc. Interspeech 2025*, pages 723–727.

Yanir Marmor, Arad Zulti, David Krongauz, Adam Gabet, Yoad Snapir, Yair Lifshitz, and Eran Segal. 2026. [VoxKneset: A large-scale longitudinal Hebrew speech dataset for aging speaker modeling](#). *Preprint*, arXiv:2603.01270.

Alec Radford, Jong Wook Kim, Tao Xu, Greg Brockman, Christine McLeavey, and Ilya Sutskever. 2023. [Robust speech recognition via large-scale weak supervision](#). In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pages 28492–28518. PMLR.

Joel Shor et al. 2019. [Personalizing ASR for dysarthric and accented speech with limited data](#). In *Proc. Interspeech 2019*, pages 784–788.

Rachael Tatman. 2017. [Gender and dialect bias in YouTube’s automatic captions](#). In *Proceedings of the First ACL Workshop on Ethics in Natural Language Processing*, pages 53–59.

Felix Weninger, Jesús Andrés-Ferrer, Xinwei Li, and Puming Zhan. 2019. [Listen, attend, spell and adapt: Speaker adapted sequence-to-sequence ASR](#). In *Proc. Interspeech 2019*, pages 3805–3809.

A Data Preparation Details

Alignment quality score. The corpus aligns each protocol to its recording with a Whisper-family

forced aligner, which assigns every protocol word a time span and a probability: how likely the model finds that word at that point in the audio. A protocol segment’s quality score is the median probability of its words. Our segments of up to 30 seconds can join several protocol segments, so their score is the duration-weighted median over those parts. A low score means the text and the audio disagree, potentially because the stenographer condensed or omitted speech, or because of cross-talk. ivrit.ai treat 0.65 as fit for training and 0.95 as high quality.

Quality filter for the analysis. The 0.7 threshold removes 7,810 of 65,990 segments (11.8%, 12.8 hours). Removing them lowers the corpus WER from 0.417 to 0.387 (A) and from 0.324 to 0.292 (B), and leaves the difference between the arms almost unchanged.

Quality filter for training. Because the score is a median, a segment can pass it while a few of its words are misaligned. Training and validation segments therefore use the word probabilities directly and must pass four conditions: confidence at least 0.95, at most 10% of words with probability below 0.5, no run of three or more such words, and the ~~number of aligned words falling within aligned words inside~~ the segment’s time span ~~matches the number of words in must account for~~ its reference text to within 20%, so that the ~~word probabilities checked above actually cover the segment’s checked probabilities cover the whole~~ text.

Two test sets. The aligner is itself a Whisper-family model, so a 0.95 threshold favours segments that Whisper models already transcribe well. ~~The score is not specific to arm B, though: per segment, it predicts arm A’s WER about as well as arm B’s (Spearman $\rho = -0.62$ and -0.63 over the 65,990 segments). It mostly measures what is hard for any model. Applied to the test set, this the threshold introduces a selection bias toward easy speech. We therefore, so we~~ score every model on two test sets drawn from the same held-out sessions: the filtered segments, and all segments with confidence of at least 0.7.

B Training Details

We ran 64 tuning runs and evaluated each one on validation loss alone~~;~~, and none of them touched the test sets. Table 1 shows the chosen settings, which are shared by every own and control adapter.

Tuning procedure. We chose four speakers for the tuning process, one per profile (30685 ~~;~~ ~~from~~ S1~~;~~ ~~23558;~~ ~~23558 from~~ S2~~;~~ ~~30718;~~ ~~30718 from~~ S3~~;~~ ~~and~~ 30859 ~~;~~ ~~from~~ S4). They were trained at the two ends of the budget range, 5 and 80 minutes~~;~~ , and 20 minutes takes the 80-minute choice.

Each run is scored by how much it lowers validation loss relative to arm B without an adapter: $(L_B - L_{\text{best}})/L_B$, where L_B is arm B’s validation loss and L_{best} is the adapter’s lowest validation loss during training (the checkpoint early stopping restores).

Tuning ran in three stages, each starting from the previous stage’s choice: the learning rate (10^{-4} , 3×10^{-4} , 10^{-3})~~;~~, then LoRA rank $\{8, 16\} \times$ Whisper’s dropout $\{0, 0.1\}$ ~~;~~ and, then augmentation (SpecAugment, alone or with $\pm 10\%$ tempo changes~~;~~, but no pitch or speed perturbation, which would change the voice).

We started from a simple default setting (learning rate 3×10^{-4} , rank 8, no dropout, no augmentation) and kept it unless a change helped clearly: a new setting replaced it only if it raised the relative drop, averaged over the four speakers, by at least 0.01, and raised it for at least three of the four.

Tuning results (Figures 8 and 9). **Learning rate matters only at 5 minutes.** ~~Only the learning rate at 5 minutes and Whisper’s dropout matter.~~ At both budgets ~~;~~ a higher rate reaches its minimum validation loss earlier~~.~~ ~~At,~~ but at 5 minutes ~~;~~ the higher rates reach a worse minimum, so we use the lowest rate, 10^{-4} . ~~At,~~ while at 80 minutes ~~;~~ ~~all three rates all three~~ reach nearly the same minimum ~~;~~ ~~so and~~ we keep the default, 3×10^{-4} . **Rank 16 does not help.** ~~It Rank 16~~ matches rank 8 at 80 minutes and is slightly worse at 5. **Augmentation does not help.** ~~The improvement is below the~~ Augmentation improves the relative drop by less than the 0.01 threshold at both budgets. **Dropout in the base model prevents learning.** Raising Whisper’s own dropout to 0.1 removes most of the adapter’s improvement for every speaker.

Beyond 80 minutes. Four speakers are trained beyond 80 minutes: 23558 and 23641 up to 1,440 minutes, and 30752 and 23635 up to 360. 23558 and 30752 are from the 12. 23641 and 23635 add profiles the 12 lack: a Haredi speaker with one of the highest WERs, whom the Hebrew fine-tune helped little, and an Ethiopian-born woman. Their data follow the same rules (Appendix A), with the newest ~~meetings sessions~~ held out for testing. The

Hyperparameter	Value
Base model	ivrit-ai/whisper-large-v3, frozen, bfloat16
Adapter	LoRA on the query and value projections of every attention layer (encoder, decoder self- and cross-attention)
Rank / α / LoRA dropout	8 / 16 / 0.05
Trainable parameters	3,932,160
Learning rate	10^{-4} at 5 minutes; 3×10^{-4} at 20 and 80 minutes
Schedule	constant after 20 warm-up steps
Batch size	8, no gradient accumulation
Weight decay	0.01
Whisper dropout	0
Augmentation	none
Training length	at most 400 steps
Validation	loss every 20 steps
Early stopping	after 4 validations without improvement; best checkpoint restored
Training labels	protocol text
Seeds	0 at every budget; 1 and 2 at 80 minutes

Table 1: Training hyperparameters, identical for every own and control adapter.

budgets are 5, 20, 80, 160 and 360 minutes, and 720 and 1,440 for the first two, each a separate adapter on a nested subset. For 23558 and 30752 the 5, 20 and 80-minute adapters are those of the main experiment. Seeds 0–2 are run at 80 minutes and at each speaker’s largest budget, and seed 0 elsewhere. Every budget has its own control. 23641 and 23635 are not in the two folds, so they are scored against the control of 23558’s fold, whose pool excludes their meetingsessions.

The recipe of Table 1 is kept, with three changes. The step cap grows with the data, to 1,200 steps at 160 and 360 minutes and 3,000 at 720 and 1,440, and no run reaches it. The learning rate was checked again at 360 minutes by the same rule, and 10^{-3} was worse than 3×10^{-4} for all four speakers (mean drop 0.381 against 0.391), so 3×10^{-4} is kept. Finally, the patience of four validations (80 steps) is about half a pass over 360 minutes of data but a tenth of a pass over 1,440, so at 720 and 1,440 minutes training may not stop before one full pass. At 1,440 minutes the best checkpoint comes after 0.83–1.11 passes, and training stops at 1.02–1.23 passes: validation loss stops improving after about one pass.

Adapter capacity. If a rank-8 adapter could not hold both the committee setting and the voice, more data could crowd out the personal part. We there-

fore repeat the 1,440-minute own adapters of 23558 and 23641 and their control at rank 32 (15.7M trainable parameters, $\alpha = 64$), seed 0. Personalization moves in both directions (Table 2), by up to 5 points, of the order of the seed-to-seed variation, and no rank-32 adapter comes near 23558’s +9.7% at 80 minutes. The own adapter itself gains no more at rank 32.

Speaker	Test	Rank 8	Rank 32
23558	full	+2.6	+4.4
23558	clean	+7.1	+2.9
23641	full	+1.5	+2.2
23641	clean	+1.7	+6.7

Table 2: Personalization (%) at 1,440 minutes, seed 0, with a rank-8 and a rank-32 adapter.

C Implementation Details

Training. Adapters are trained with PyTorch, Hugging Face transformers (Seq2SeqTrainer) and peft, on the frozen base model in bfloat16. Training segments are fed latest meeting-session first, so the 5-, 20- and 80-minute budgets are nested prefixes of one ordering. Each control’s pool takes one segment from each of its six speakers in turn until the budget is reached, and never from a session in which a speaker it is scored on is tested or validated.

Scoring. Reference and hypothesis are normalized (niqqud and punctuation removed) and aligned word by word; and substitutions, deletions and insertions are counted per segment. WER is pooled over a speaker’s test set: the sum of word errors divided by the sum of reference words. Decoding is greedy, and every transcription passes a loop guard: a hypothesis whose zlib compression ratio exceeds 3.0 is decoded again with no 6-token sequence repeated, and replaced if the new one compresses less, identically for the base model, own adapters and controls. Confidence intervals and p-values come from a paired bootstrap over test segments (~~2,000 resamples, two-sided p~~); ~~for personalization, the own and control hypotheses of the same segments are resampled together~~ Appendix E).

D Additional Results

D.1 Speaker-Level Error Map

Reliability of the per-speaker measures. Each speaker is measured on about one hour of speech,

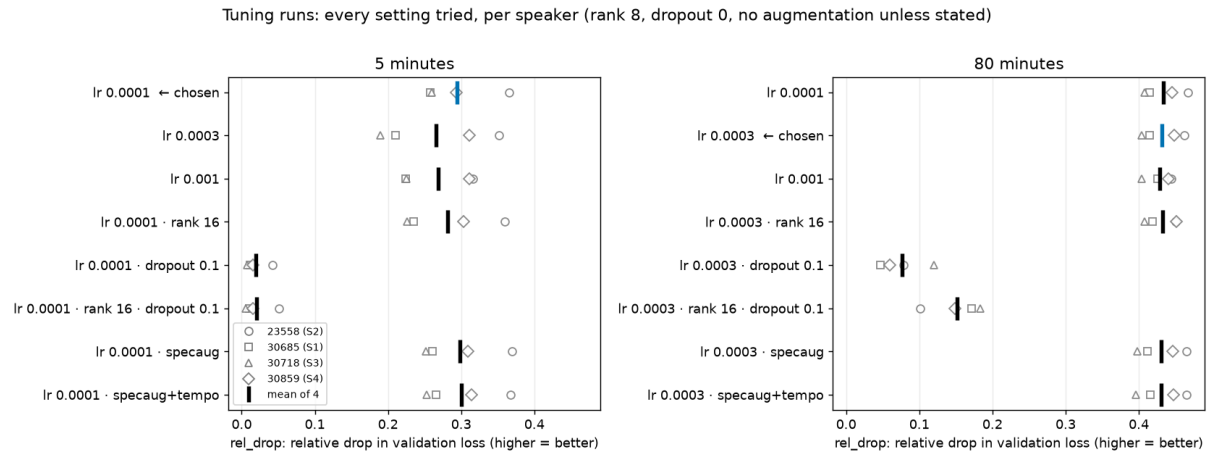


Figure 8: Every tuning run, at 5 minutes (left) and 80 minutes (right). Markers are the four tuning speakers, the bar is their mean, and blue is the chosen setting. The x-axis is the relative drop in validation loss from arm B to the best checkpoint.

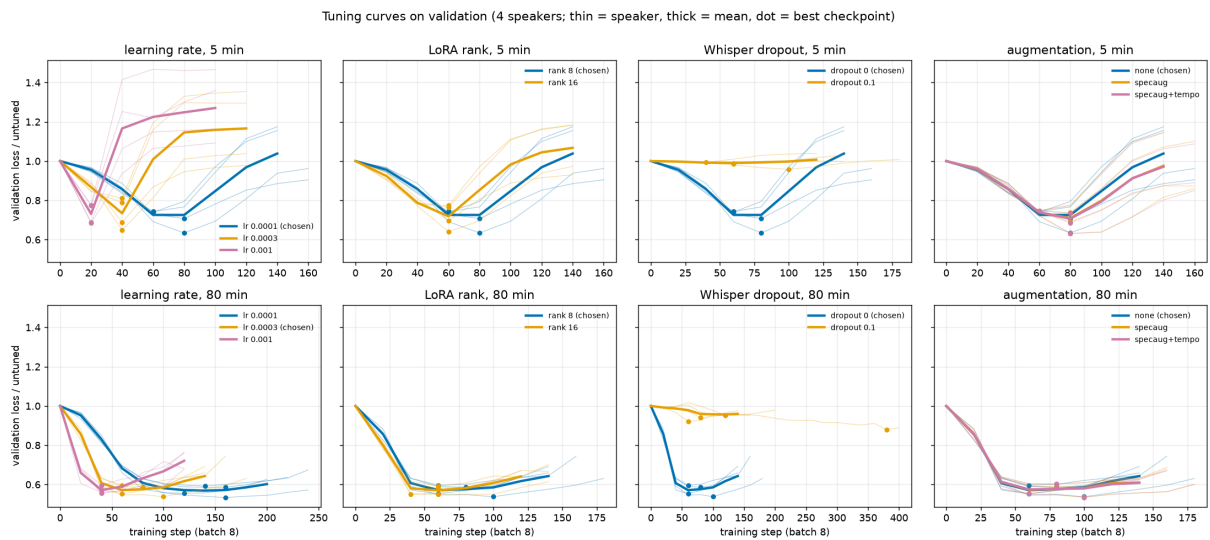


Figure 9: Validation loss relative to arm B during tuning, varying one setting at a time around the chosen recipe, at 5 minutes (top) and 80 minutes (bottom). Thin lines are the four speakers, the thick line is their mean, and dots are the best checkpoints.

so their WER and gain carry sampling noise. To test whether the differences between speakers are a stable property of the speakers rather than of the particular sample, we split each speaker’s sessions into two alternating halves, compute WER_B and the relative gain on each half independently, and compare the halves across speakers (Figure 11). If the differences were noise, the two halves would be unrelated ~~and the points would form a shapeless cloud~~. If, and if they are stable, the points follow the diagonal. They do: the correlation between halves, corrected to full length (Spearman–Brown), gives a reliability of 0.78 for WER_B and 0.75 for relative gain. This is what justifies ranking speakers by their gain when selecting them for personaliza-

tion.

Gain by group. Figure 12 shows the gain for the six groupings not shown in the main text (speaking rate is Figure 2). Speakers with missing metadata and groups of fewer than three speakers are omitted.

Error composition. Over half of arm B’s errors are insertions (Table 3): words in the transcript that the protocol does not contain. All WERs in this paper are computed against the protocol as is.

Recording conditions. WER also varies with the recording conditions: arm B scores 0.32–0.37 in 2018–2019 and 0.27–0.29 from 2020 on (Fig-

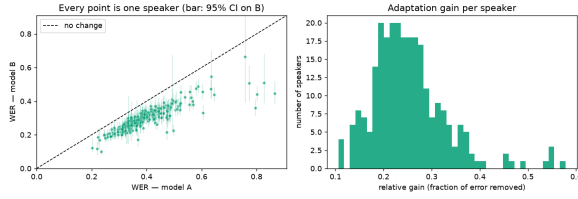


Figure 10: Left: per-speaker WER under arm A against arm B. Every point is one speaker, the bar is the 95% bootstrap interval on WER_B and the dashed line is no change. Right: distribution of the relative gain over the 261 speakers with at least 20 segments.

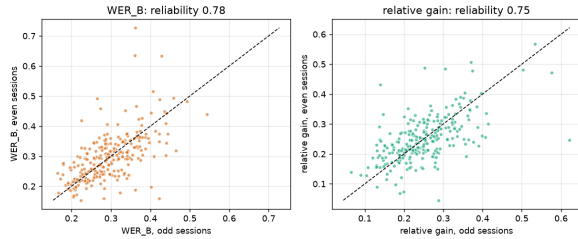


Figure 11: Each point is one speaker, measured separately on their odd and even sessions. Points near the diagonal mean the measure repeats across recordings, and titles give the split-half reliability (Spearman–Brown corrected).

ure 13), and 0.21–0.37 across the twelve largest committees. Personalization splits are therefore ~~meeting-disjoint~~ session-disjoint and ordered by date: each speaker is adapted on earlier ~~meetings~~ sessions and tested on their newest ones, which is also how a personalized model would be used in practice.

D.2 Personalization

Up to 360 minutes. Figure 17 shows the two speakers trained on up to 360 minutes. As for the speakers trained on 24 hours, the own adapter keeps improving (on the full test, from +23.3% to +26.9% for 30752 and from +15.0% to +19.5% for 23635), the control improves as fast, and personalization at 360 minutes is no higher than at 80 (30752 from +4.7% to +1.7%, 23635 from +0.4% to −0.1%).

Decision rule. Before the longer runs we fixed a rule on the full test, comparing each speaker’s personalization at the largest budget with that at 80 minutes: ~~personalization grows with speaker~~ Personalization would count as growing with data if the difference is were at least +2 points , with its interval above zero , for at least two of the four speakers, and ~~it does not as not growing~~

Model	WER	CER	share of word errors		
			Sub.	Del.	Ins.
A (general)	0.387	0.241	49%	13%	38%
B (Hebrew fine-tune)	0.292	0.203	36%	11%	53%

Table 3: Corpus error rates after the quality filter (267 speakers, 58,180 segments) and the composition of each model’s word errors: substitutions, deletions and insertions.

if every interval ~~contains~~ contained zero with an upper end below +2. Neither holds. ~~No interval lies above zero, and for three speakers the interval contains zero with an upper end below +2, but for~~ : three intervals meet the second condition, but 23558 it lies’s lies entirely below zero (−5.6, [−7.7, −3.6]). ~~By its wording the~~ The rule is therefore undecided by its wording, while the ~~direction of the result is against personalization growing with speaker data~~ result points against growth.

E Significance Testing

Every comparison in Section 4.2 is paired: two models transcribe the same test segments of one speaker, so we ask whether one makes fewer errors than the other on those segments.

What is compared. For the gain of an adapter over arm B, the difference is $WER_B - WER_{\text{adapted}}$. For personalization, it is $WER_{\text{control}} - WER_{\text{own}}$: the own adapter and the control on the same segments. Both are pooled differences, the sum of each model’s word errors over the segments divided by the sum of reference words.

Resampling. We draw as many segments as the speaker’s test set holds, with replacement, and recompute the pooled difference on the draw; ~~this~~ . This is repeated 2,000 times. The same draw is applied to both models, so a hard segment counts for both or for neither, and the variation between segments largely cancels. We resample whole segments rather than words because errors within a segment are not independent: a misheard name or a runaway transcription produces several at once.

Interval and p-value. The 95% interval is the 2.5th to 97.5th percentile of the 2,000 differences, in absolute WER; ~~and~~ dividing by WER_B gives it in the relative units of Section 4.2. The two-sided p-value is twice the smaller of the shares of draws at or below zero and at or above zero, so it resolves

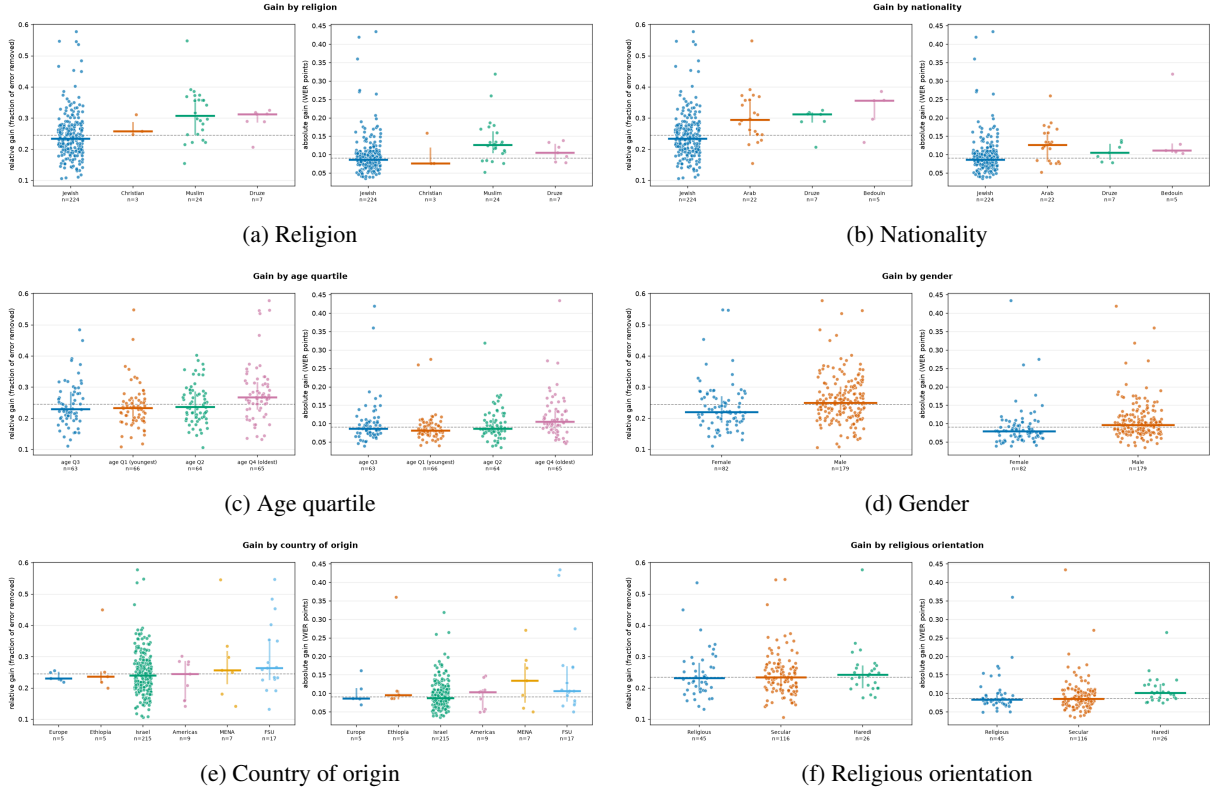


Figure 12: Gain from the Hebrew fine-tune by group, relative (left of each panel) and absolute (right). Each point is one speaker with at least 20 segments. The bar is the group median, the line the interquartile range, and the dashed line the overall median.

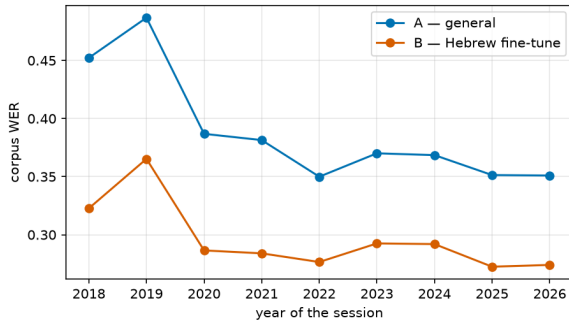


Figure 13: Corpus WER of both models by the year of the session, after the quality filter.

p-values down to 0.001; and smaller ones are reported as $p < 0.001$.

Multiple comparisons. Each speaker–seed pair is tested on its own, without correction. If own adapters and controls did not differ, about two of 36 pairs would fall below $p = 0.05$ by chance, and about one of them with a positive sign; we. We observe 8 (clean test) and 16 (full test) significantly positive pairs and none significantly negative. Per-speaker claims rest on agreement across all three seeds rather than on a single p-value.

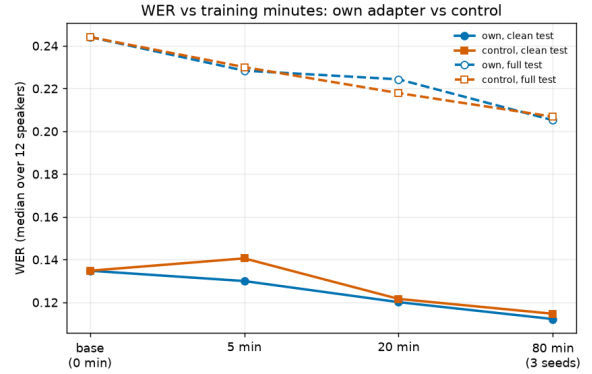


Figure 14: Median WER over the 12 speakers at 0 (arm B), 5, 20 and 80 minutes of training audio, for the own adapter (blue) and the control (orange), on the clean test (solid) and the full test (dashed). At 80 minutes, each speaker’s WER is averaged over three seeds.

F Bootstrap for the Speaker-Level Error Map

The per-speaker intervals of Section 4.1, and the bars in Figure 10, come from a simpler bootstrap than Appendix E’s, run once per speaker on arm A’s and arm B’s transcriptions of that speaker’s one-hour sample.

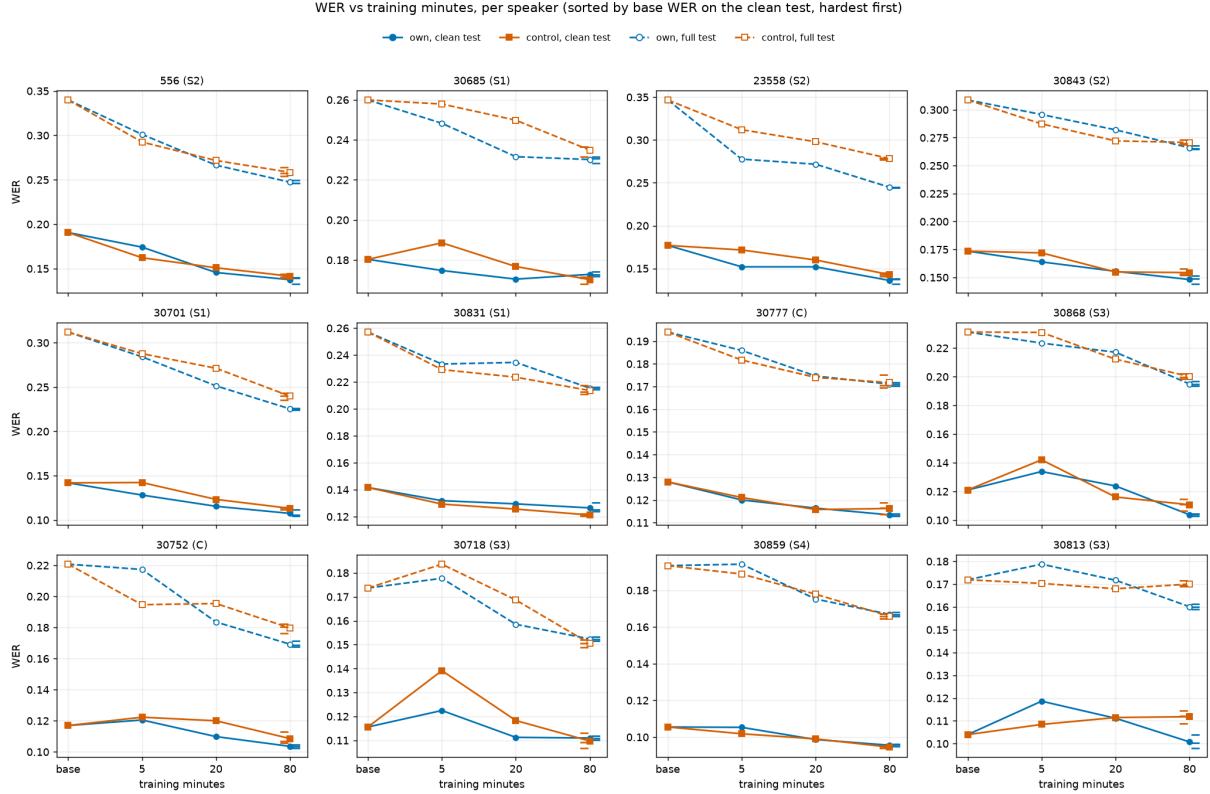


Figure 15: The same per speaker, sorted by arm B’s WER on the clean test, hardest first. Ticks at 80 minutes: the individual seeds.

Resampling. The speaker’s segments are drawn with replacement, as many as the sample holds, 1,000 times. The same draw is used for both arms. Each draw gives a pooled WER_A and WER_B (the sum of word errors over the drawn segments divided by the sum of reference words) and their difference, the absolute gain $WER_A - WER_B$.

Intervals. The 95% interval of WER_A , WER_B and the gain is the 2.5th to 97.5th percentile of the 1,000 values. No p-value is computed: a speaker’s improvement under arm B is called beyond sampling noise when the interval of their gain lies entirely above zero, as it does for all 261 speakers with at least 20 segments.

Difference from Appendix E. The resampling is the same (whole segments, paired across the two models), but it uses 1,000 draws instead of 2,000, compares the two base models instead of adapters, and reports intervals only.

G Division of Work

Both authors shaped the project from the start. We developed the idea together over several brainstorming meetings, refining a general direction into

the final research question. We then sharpened it in discussions with the ivrit.ai team, who pointed us to directions and problems they had encountered and considered worth studying.

We set the guidelines of each stage together and divided the detailed work between us. For the speaker-level analysis (Stage 1), we jointly defined the speaker groupings and the analyses to run, and Hadas Yonat, a data engineer, carried out most of the in-depth analysis. For personalization (Stage 2), we jointly designed the main principles of the training architecture, and Dolev Abudi led the training itself. Throughout, we worked in parallel and in close coordination: in regular joint working sessions, in person and over Zoom, we shared our progress and findings and made the project’s decisions together. We enjoyed working together, and this teamwork was an important and meaningful part of the project for both of us.

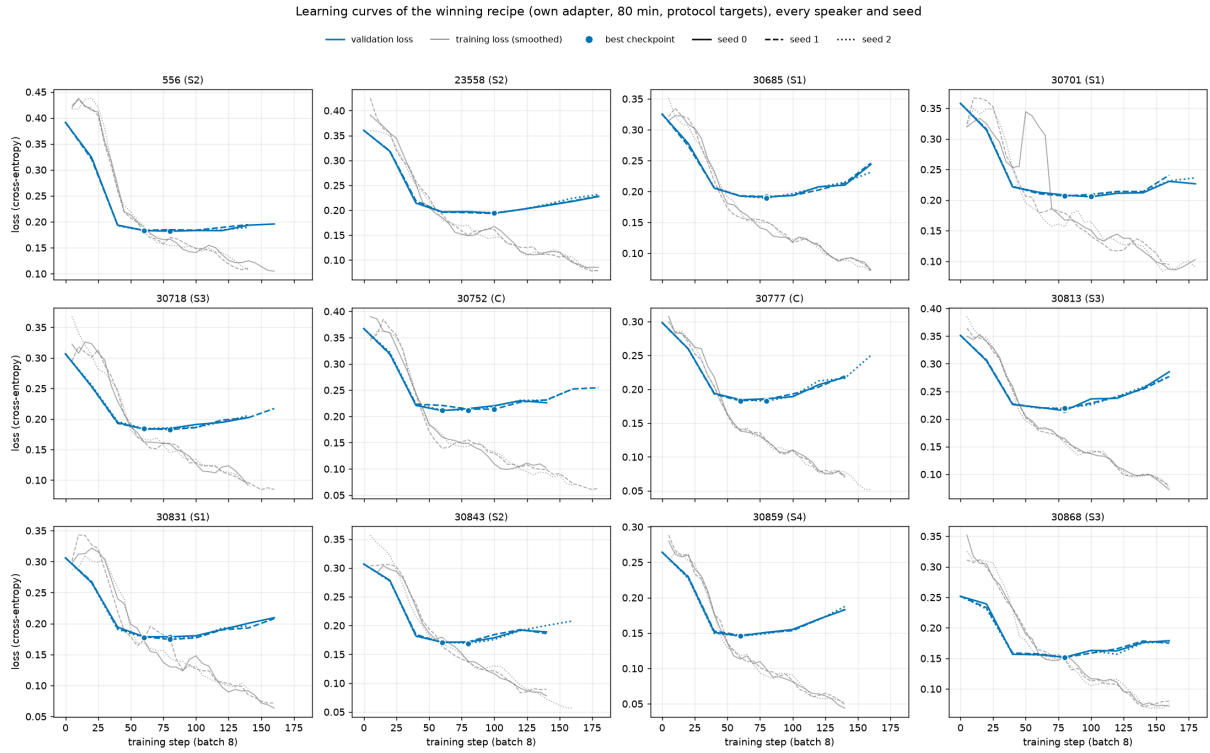


Figure 16: Learning curves of the 80-minute own adapters, one panel per speaker. Blue $\pm$ is the validation loss every 20 steps (step 0 is arm B) $\pm$, grey $\pm$ the training loss $\pm$, smoothed over 20 steps $\pm$, line style $\pm$ the seed $\pm$, and the dot $\pm$ the checkpoint early stopping restored.

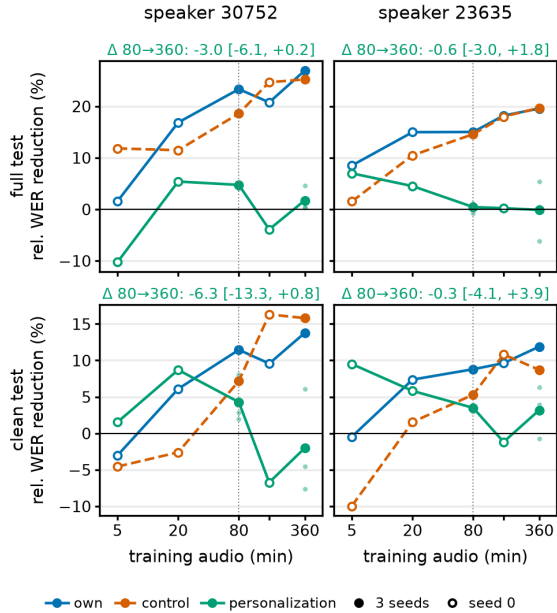


Figure 17: As Figure 6, for the two speakers trained on up to 360 minutes.